

Conflict-Induced Displacement and Settlement Change in Borno State, Nigeria (2015-2023)

*A multi-indicator geospatial assessment of settlement contraction and change
signals*

Final Validated Technical Report

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Executive Summary

This project examined settlement change in Borno State between 2015 and 2023. I developed the analysis to identify places where several independent datasets point to physical contraction, reduced activity or other forms of settlement change. The work was designed as a screening and validation exercise, not as a method for declaring settlements abandoned from satellite data alone.

I screened 2,455 settlements across all 27 local government areas (LGAs) of Borno State. The analysis brought together settlement locations, built-up change, night-time lights, conflict exposure and modelled population change. Thirteen locations were retained for detailed review because they showed combinations of signals that required closer interpretation.

The final validation produced three priority contraction cases: Mitile in Abadam, and Johana and Wumbi in Kala Balge. These locations combine strong built-signal decline with modelled population decline. Seven other settlements showed strong physical, activity or conflict-related change signals, but their modelled population was stable or increasing. I therefore retained them as priority change-signal cases rather than treating them as abandonment cases. Population evidence was insufficient for the remaining three locations.

The most important result is also a limitation: none of the 13 detailed cases can be described as confirmed abandonment from the available datasets. The evidence is strong enough to identify settlement contraction and locations that deserve further investigation, but confirmation of abandonment would require direct field evidence or another reliable source that establishes occupancy status.

Project measure	Validated result
Study period	2015-2023
Settlements screened	2,455
LGAs covered	27
Detailed validation cases	13
Priority contraction cases	3
Priority change-signal cases	7
Cases with insufficient population evidence	3
Confirmed abandoned settlements	0

1. Introduction

Borno State has experienced prolonged conflict and large-scale population displacement. These changes can affect settlements in different ways: people may leave temporarily or permanently, buildings may be damaged or disappear, economic activity may reduce, and some places may later recover or receive new residents. A single dataset cannot describe all of these processes reliably.

I therefore approached the project as a settlement-change assessment. Instead of assuming that a decline in one indicator meant abandonment, I compared several forms of evidence and used them together to identify locations where the signals were consistent, contradictory or incomplete. This became particularly important during validation, because some settlements with strong built-up or night-light decline also showed stable or increasing modelled population.

1.1 Aim

The aim was to identify and assess settlement contraction and other change signals across Borno State between 2015 and 2023 using multiple geospatial indicators.

1.2 Objectives

- Screen settlements across Borno State for measurable signs of physical or activity decline.
- Assess settlement-level exposure to recorded conflict events.
- Compare built-up, night-light and population evidence rather than relying on one indicator.
- Identify a small set of locations for detailed validation.
- Separate strong contraction evidence from change signals that do not support an abandonment claim.
- Produce clear maps, rankings and a final scientific register that can support further field investigation.

2. Study Area

Borno State is located in north-eastern Nigeria and contains 27 LGAs. The statewide screening covered 2,455 populated places represented in the settlement dataset. The final 13 detailed validation cases were located in Abadam, Bama, Gwoza, Kala Balge, Konduga and Ngala LGAs.

The final case distribution was not intended to represent the whole state statistically. These 13 places were selected because their evidence profiles made them useful for detailed interpretation after the statewide screening.

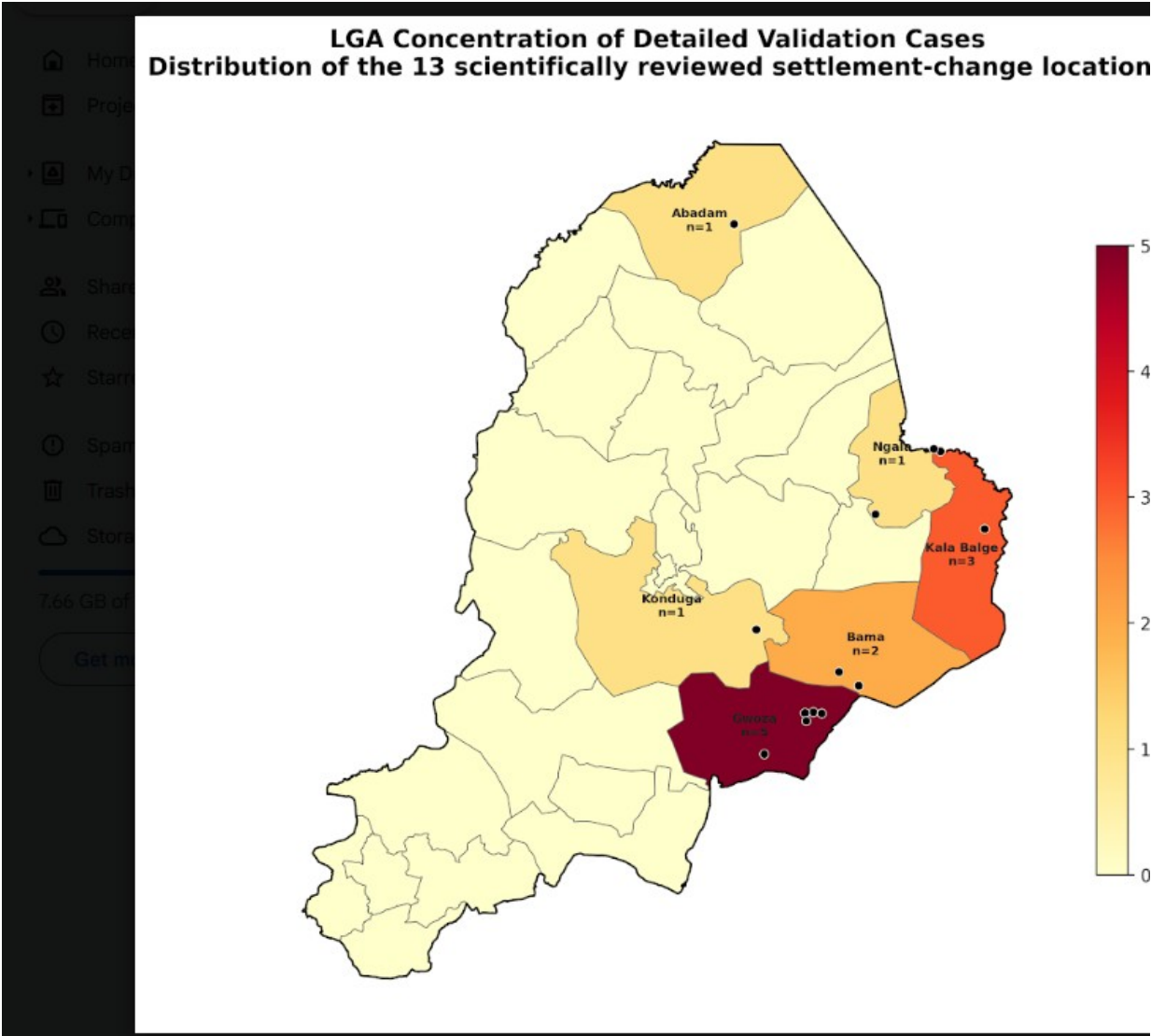


Figure 1. LGA concentration of the 13 detailed validation cases.

3. Data and Evidence

I combined datasets that describe different parts of settlement change. Settlement points provided the analysis units. Satellite-derived built-up information was used to assess physical change. Night-time lights provided an additional indication of activity change. Conflict records described exposure to recorded events, while WorldPop estimates were used as demographic evidence. Administrative boundaries were used to place the results within Borno State and its LGAs.

Evidence	Main source used in the project	Role in interpretation
Settlement locations	GeoNames populated places	Statewide settlement screening and case identification
Conflict exposure	ACLED event records	Settlement-level conflict

		context
Night-time activity	VIIRS night-time lights	Detection of persistent or temporally consistent light decline
Built environment	Satellite-derived built-up change, including Dynamic World outputs	Physical built-signal decline
Population	WorldPop modelled population	Demographic corroboration, contradiction or insufficient evidence
Administrative boundaries	Borno State and LGA boundaries	Spatial context and LGA summaries

These datasets do not measure the same thing. A fall in night-time light is not the same as population loss, and built-signal decline is not direct proof that nobody lives in a settlement. I therefore treated agreement between indicators as stronger evidence than any single result.

4. Method

4.1 Statewide screening

I began with 2,455 settlements and calculated or attached evidence for built-signal change, night-light change, conflict exposure and population change. The screening created a common settlement-level evidence table so that the indicators could be compared consistently.

4.2 Multi-indicator interpretation

The indicators were used as evidence flags rather than as direct abandonment labels. Built-signal decline described physical change. Night-light decline described reduced detected activity. Conflict exposure provided context. Population decline strengthened a contraction interpretation, while stable or increasing population weakened any literal abandonment claim.

4.3 Detailed validation

Thirteen settlements were retained for detailed review. For each case, I compared the evidence profile and checked whether the indicators supported one another. This stage was important because the strongest-looking decline signal was not always supported by the population evidence.

4.4 Final classification

The final classes describe the strength and consistency of the evidence. A1 identifies strong physical-demographic contraction. B1 identifies strong decline signals where modelled population increased. B2 identifies strong decline signals where population persisted. C1 identifies a settlement decline signal where the population evidence was insufficient.

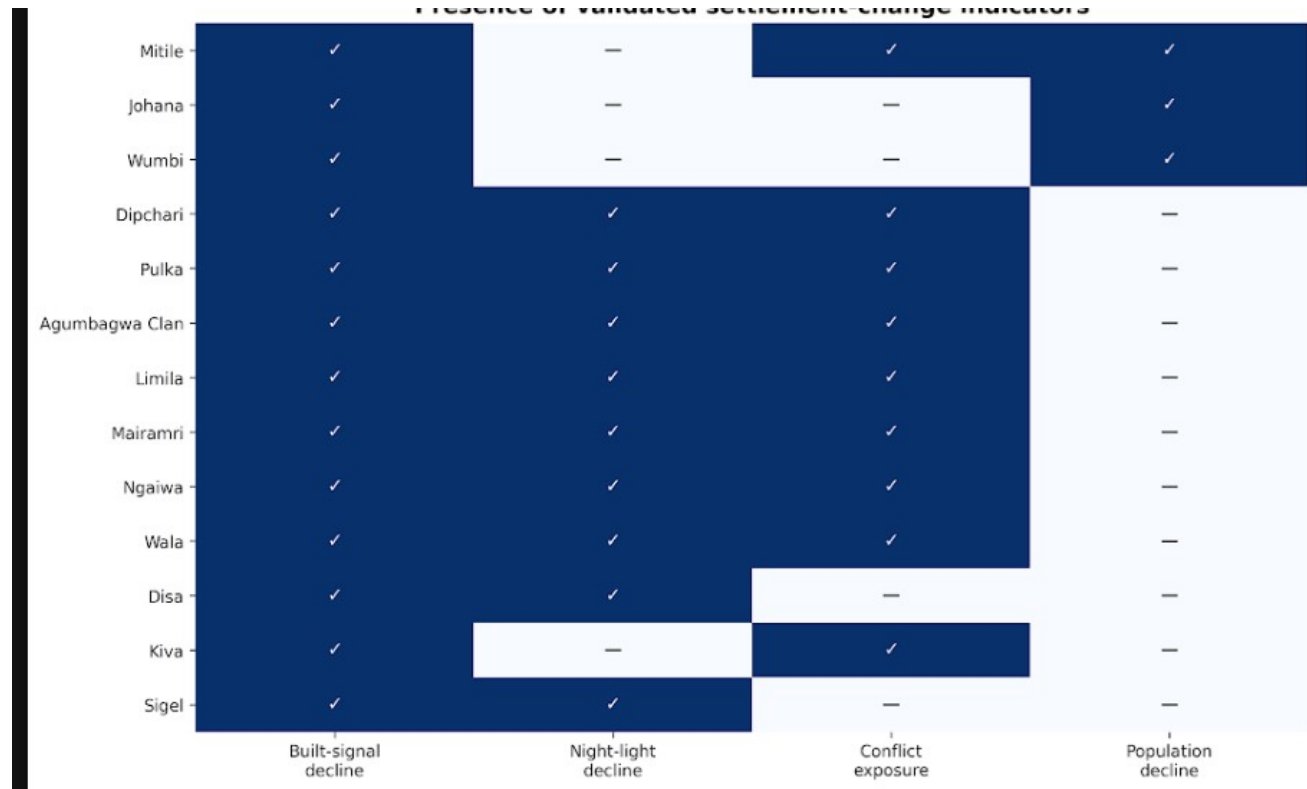


Figure 2. Evidence profile of the 13 detailed validation cases.

5. Results

5.1 Statewide evidence pattern

The statewide synthesis showed that most settlements did not display substantial settlement-contraction evidence. Of the 2,455 settlements, 1,368 fell in the no-substantial-contraction group. Other locations showed individual or combined indicators, including built-signal decline, conflict exposure, night-light decline and population decline. This wider screening was used to narrow the analysis to cases where several signals required closer attention.

The audit reproduced the four-indicator evidence count for all 2,455 records without mismatch. Eight settlements had three of the four main evidence indicators, 228 had two, 851 had one and 1,368 had none. The final detailed group was not selected simply by taking the highest count; the interpretation also considered what each indicator meant and whether population evidence supported the physical signal.

5.2 Final classification of the 13 detailed cases

The final classification contains three A1 cases, four B1 cases, three B2 cases and three C1 cases. This distribution is important because it shows that the detailed locations do not represent one uniform type of settlement change.

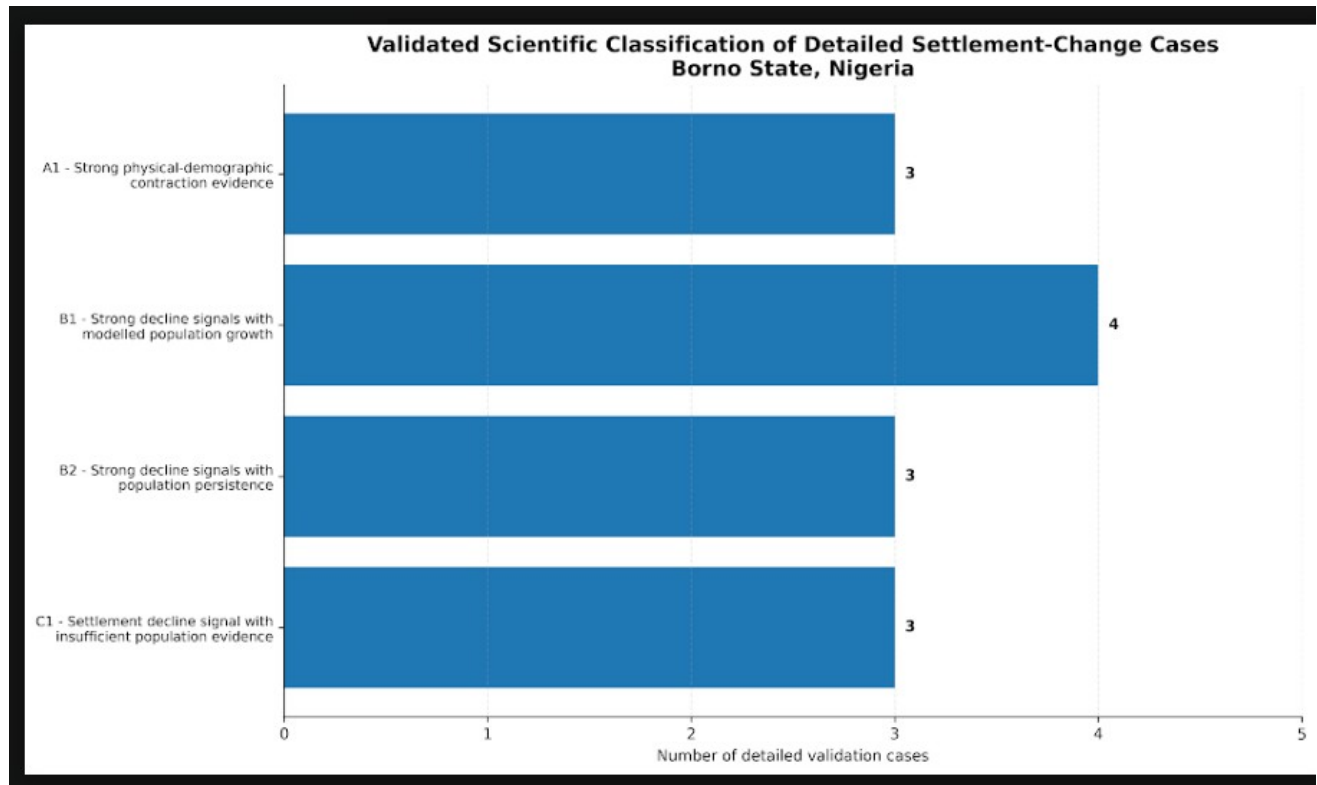


Figure 3. Final scientific classification of the detailed settlement-change cases.

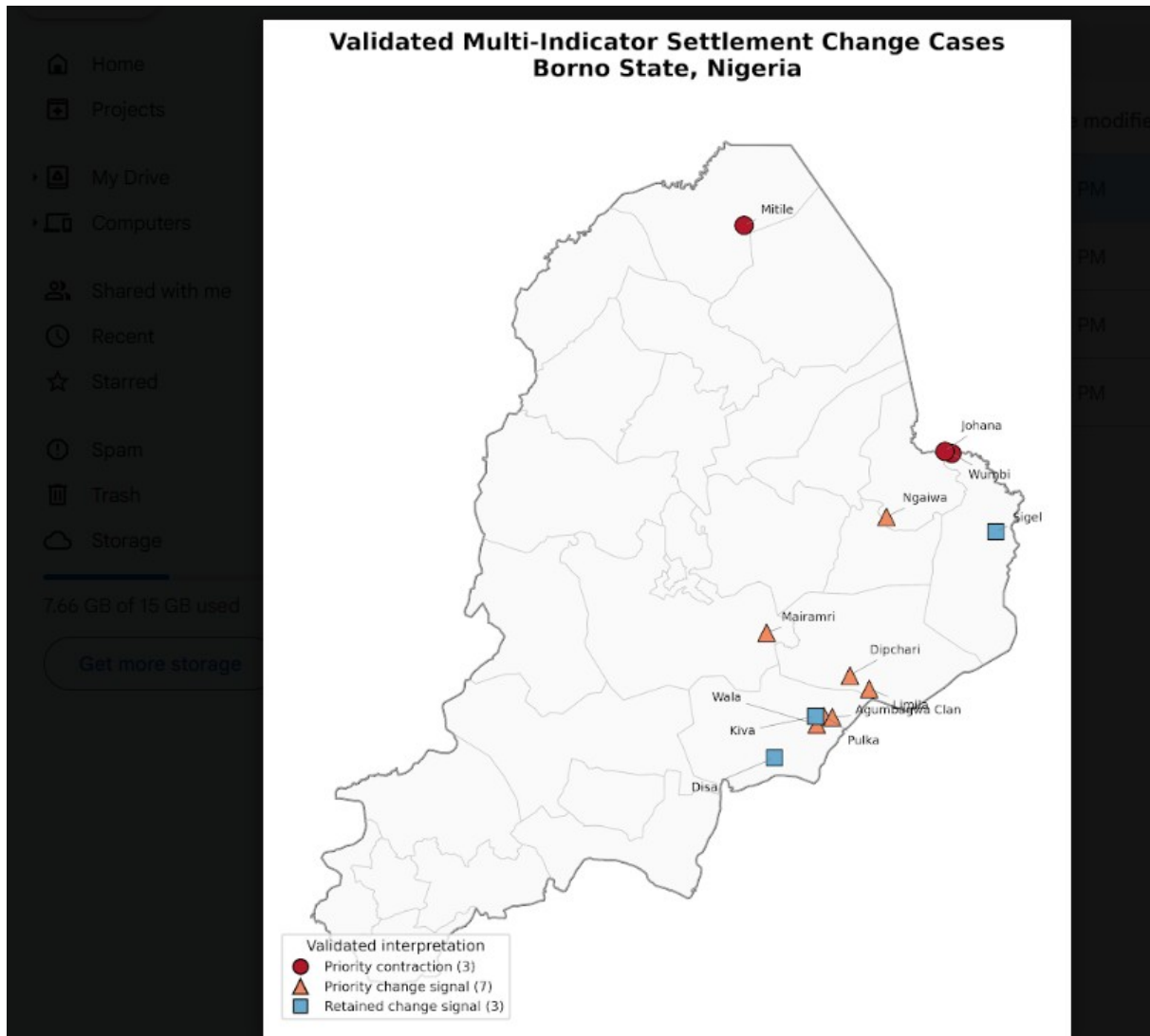


Figure 4. Spatial distribution of the final validated interpretations.

5.3 Priority contraction cases

Mitile, Johana and Wumbi form the strongest contraction group. In all three locations, strong built-signal decline is accompanied by modelled population decline. This agreement provides physical and demographic support for a contraction interpretation.

Mitile received the highest validated support score of 7. Johana and Wumbi each scored 6. I treat these as priority locations for follow-up because the evidence is more internally consistent than in the other detailed cases. Even here, the result should be described as strong settlement contraction evidence rather than confirmed abandonment.

5.4 Change signals contradicted by population evidence

Seven locations showed strong combinations of built, night-light or conflict-related evidence while the population model indicated persistence or growth. These were Dipchari, Pulka, Agumbagwa Clan, Limila, Mairamri, Ngaiwa and Wala.

Four of these settlements - Dipchari, Limila, Ngaiwa and Wala - had modelled population increases. Pulka, Agumbagwa Clan and Mairamri had relatively stable population. The physical and activity

signals remain useful, but they do not support a simple abandonment interpretation. These places may reflect partial contraction, spatial restructuring, damage followed by recovery, limitations in the population model, or other local processes that cannot be separated with the available data.

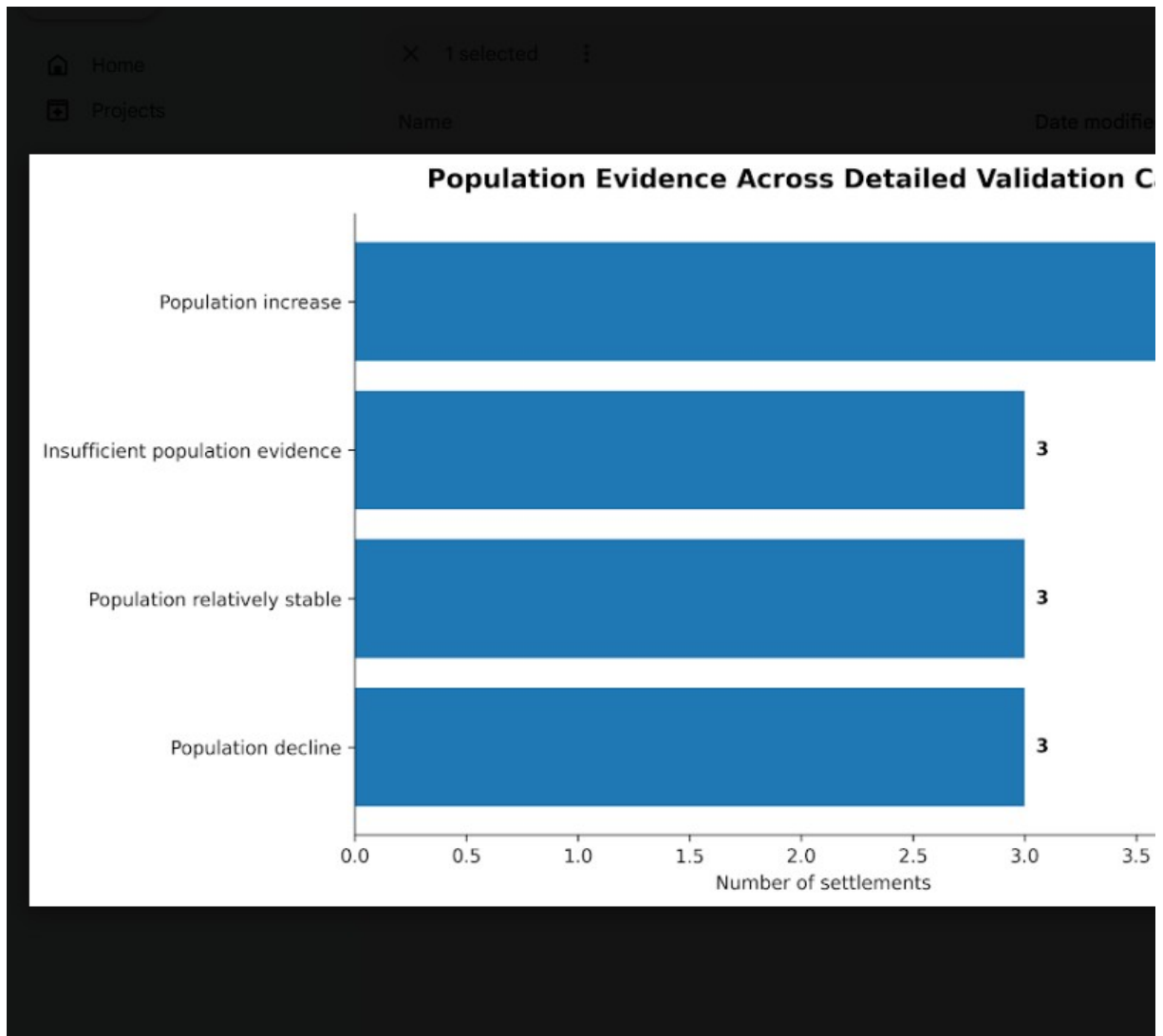


Figure 5. Population evidence used to corroborate or limit settlement-change interpretations.

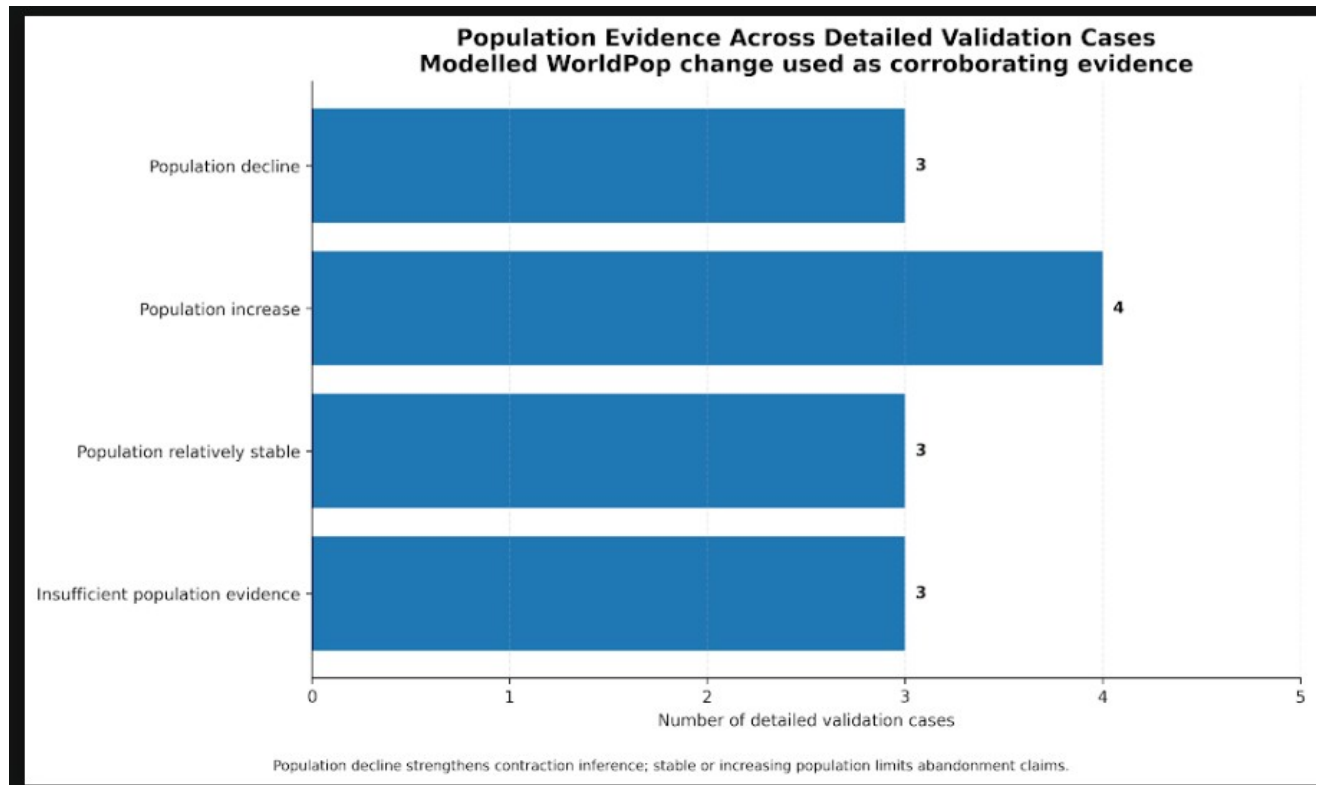


Figure 6. Distribution of population evidence across the 13 detailed cases.

5.5 Cases with insufficient population evidence

Disa, Kiva and Sigel retained settlement-change signals, but the available population evidence was insufficient. I therefore kept them in the final register as change-signal cases without using demographic evidence to strengthen or reject the interpretation.

5.6 Scientific support ranking

The support score summarises the validated evidence available for each detailed case. It is a synthesis priority measure, not a probability that a settlement is abandoned. Mitile ranked first, followed by Johana and Wumbi. The ranking helps organise follow-up work but should always be read together with the scientific class and population interpretation.

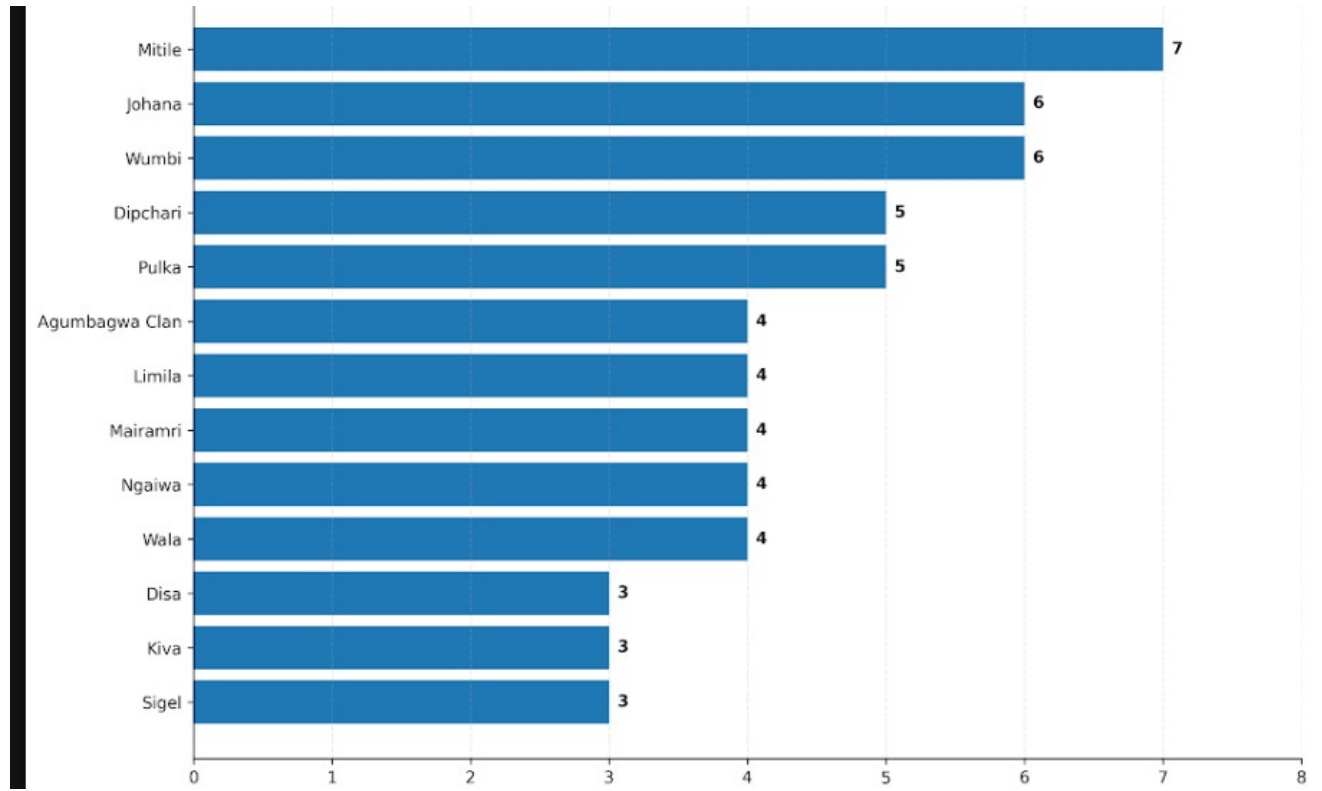


Figure 7. Validated multi-indicator scientific support ranking.

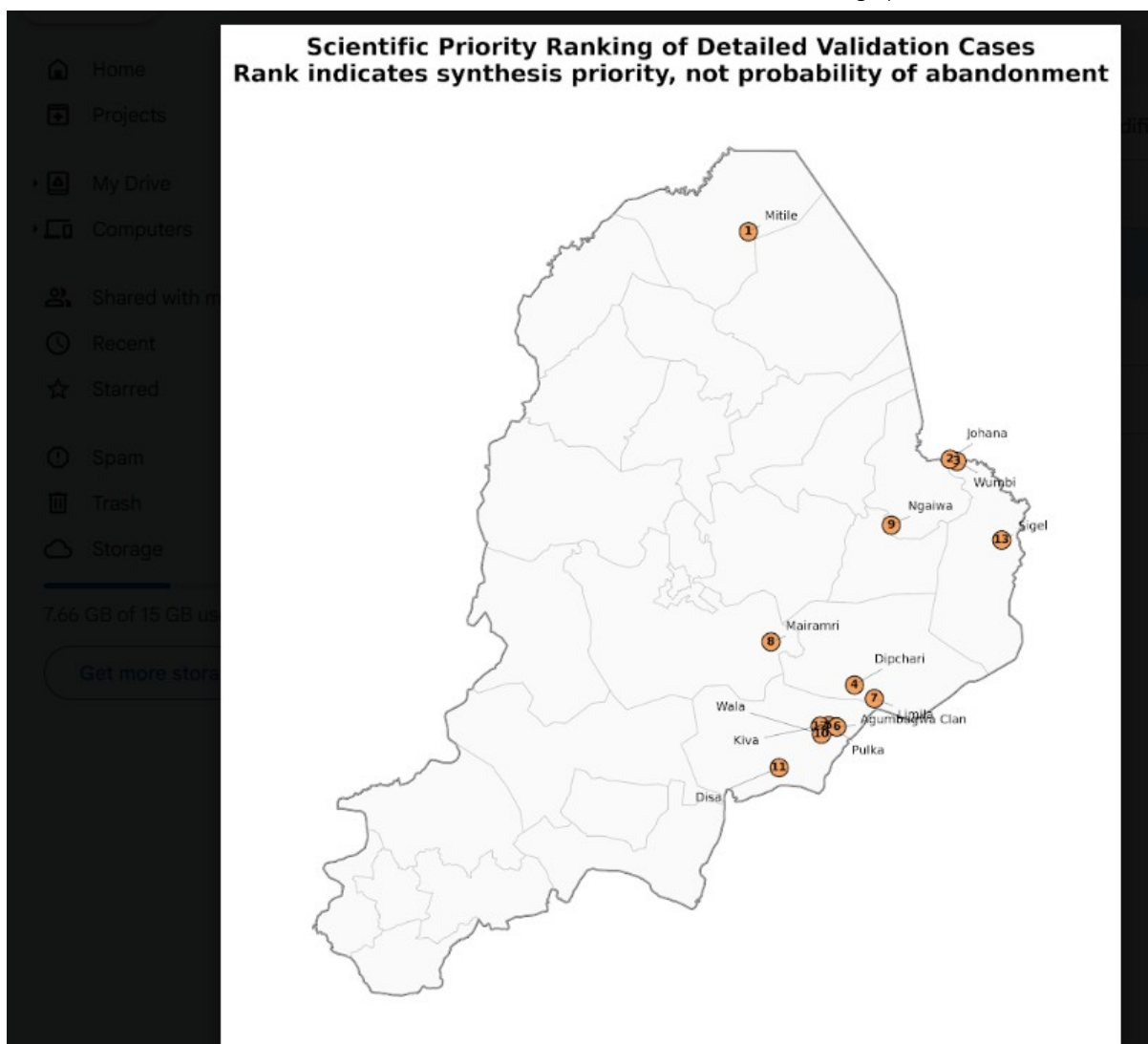


Figure 8. Spatial distribution of the scientific priority ranking.

6. Detailed Validation Register

The table below records the final interpretation used in this report. The wording deliberately separates contraction evidence from abandonment.

Rank	Settlement	LGA	Score	Class	Public interpretation
1	Mitile	Abadam	7	A1	Priority contraction case
2	Johana	Kala Balge	6	A1	Priority contraction case
3	Wumbi	Kala Balge	6	A1	Priority contraction case
4	Dipchari	Bama	5	B1	Priority change-signal case
5	Pulka	Gwoza	5	B2	Priority change-

Rank	Settlement	LGA	Score	Class	Public interpretation
					signal case
6	Agumbagwa Clan	Gwoza	4	B2	Priority change-signal case
7	Limila	Bama	4	B1	Priority change-signal case
8	Mairamri	Konduga	4	B2	Priority change-signal case
9	Ngaiwa	Ngala	4	B1	Priority change-signal case
10	Wala	Gwoza	4	B1	Priority change-signal case
11	Disa	Gwoza	3	C1	Retained change-signal case
12	Kiva	Gwoza	3	C1	Retained change-signal case
13	Sigel	Kala Balge	3	C1	Retained change-signal case

7. Interpretation and Planning Meaning

The project shows why settlement change in a conflict-affected region should not be reduced to a single map of 'abandoned' places. Different indicators can move in different directions. A settlement can lose built-up signal or night-time activity while a population model still indicates persistence or growth.

For planning and humanitarian work, the most useful output is therefore a hierarchy of places that deserve different forms of follow-up. The three priority contraction cases deserve the closest field or high-resolution review. The seven population-contradicted cases deserve investigation into the reason for the mismatch. The three cases with insufficient population evidence require better demographic information before stronger conclusions are made.

- Use the priority contraction cases to guide targeted field verification and high-resolution image review.
- Do not use the final 13 as a list of abandoned settlements.
- Combine settlement-change evidence with displacement records, local knowledge and current occupancy information before making operational decisions.
- Repeat the analysis when newer population, conflict and satellite data become available so that recovery and renewed decline can be distinguished.
- Keep the settlement-level evidence register alongside the maps so that decision-makers can see why each location received its classification.

8. Limitations

The analysis is limited by the meaning and resolution of the source datasets. WorldPop is a modelled population surface rather than a direct census of each settlement. It can therefore support or challenge an interpretation, but it cannot independently confirm occupancy.

Night-time lights are affected by electricity access, sensor characteristics, settlement size and other factors. Reduced light does not automatically mean population loss.

Satellite-derived built-up change describes changes in the detected built signal. It does not directly establish whether buildings are occupied, damaged, demolished or temporarily unused.

Conflict exposure shows recorded events near settlements, but spatial association does not prove that a particular settlement change was caused by a specific event.

The final 13 locations are detailed validation cases selected from the wider screening. They should not be treated as a statistically representative sample of all settlements in Borno State.

Most importantly, the available datasets do not directly establish abandonment. The project therefore reports zero confirmed abandoned settlements.

9. Conclusion

I screened 2,455 settlements across Borno State and used several independent indicators to examine settlement change between 2015 and 2023. The detailed validation retained 13 locations, but the final evidence did not support treating all of them as abandoned settlements.

Three settlements - Mitile, Johana and Wumbi - show the strongest physical-demographic contraction evidence. Seven locations show important physical or activity decline signals that are not supported by population loss, while three retain change signals with insufficient population evidence. These differences are central to the final interpretation.

The main contribution of the work is a transparent way to move from broad geospatial screening to a smaller set of carefully interpreted cases. The results can help direct further investigation, but field evidence remains necessary before making claims about actual abandonment or the cause of settlement change.

10. Data Sources and Project References

- ACLED. Armed Conflict Location & Event Data used for settlement-level conflict exposure.
- GeoNames. Populated-place records used to define the statewide settlement universe.
- WorldPop. Modelled population data used to assess population change between 2015 and 2023.
- VIIRS night-time lights. Monthly satellite observations used to assess temporally consistent changes in detected night-time activity.
- Dynamic World / satellite-derived land-cover outputs. Used as part of the built-signal change assessment.
- Borno State and LGA administrative boundary datasets used for spatial context and LGA summaries.

- Project final scientific register: Borno_FINAL_VALIDATED_Settlement_Scientific_Register.
- Project validated findings register: Borno_FINAL_VALIDATED_Project_Findings.

Appendix A. Final Classification Definitions

Class	Meaning	How it is used
A1	Strong physical-demographic contraction evidence	Strong built-signal decline and modelled population decline converge. Priority contraction case; not confirmed abandonment.
B1	Strong decline signals with modelled population growth	Physical/activity decline signals are present, but modelled population increased. Abandonment interpretation is not supported.
B2	Strong decline signals with population persistence	Physical/activity decline signals are present while modelled population remains relatively stable. Abandonment interpretation is not supported.
C1	Settlement decline signal with insufficient population evidence	Change signal retained, but demographic evidence is insufficient for a stronger interpretation.